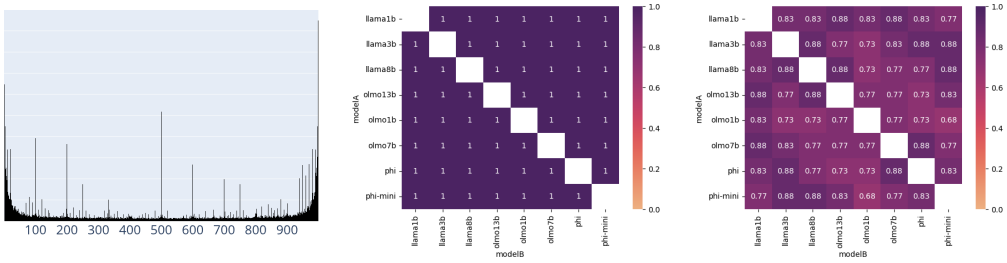


(a) Contributions of Fourier base frequencies (Llama-3 1B, provided as an example)

(b) IoU for number pieces

(c) IoU for random pieces

Figure 2: Intersection-over-union of top $k = 63$ Fourier base frequencies

3.2 Number Representations are Always Sinusoidal

In subsection 3.1, we presented evidence that the input embeddings of numbers in LLMs have certain properties related to how they encode numeric information, such as being sinusoidal. In this section, we aim to find whether such representations are also maintained and employed throughout the models.

For qualitative assessment, we visualize the PCA and its Fourier transform of internal activations of numerical tokens in Section A.3. Besides the apparent wave-like pattern, the representations have a sparse Fourier transform, confirming the sinusoidal character. For quantitative analysis, we look at the model's internal representations through the lens of the sinusoidal probe proposed by Kadlčík et al. (2025). This probe was designed to map input embeddings of LLMs into an integer, thus classifying embeddings into predefined range of numeric values. The sinusoidal probe is defined as:

$$f_{\sin}(\mathbf{x}) = (\mathbf{W}_{\text{out}}\mathbf{S})^T(\mathbf{W}_{\text{in}}\mathbf{x}) \tag{1}$$

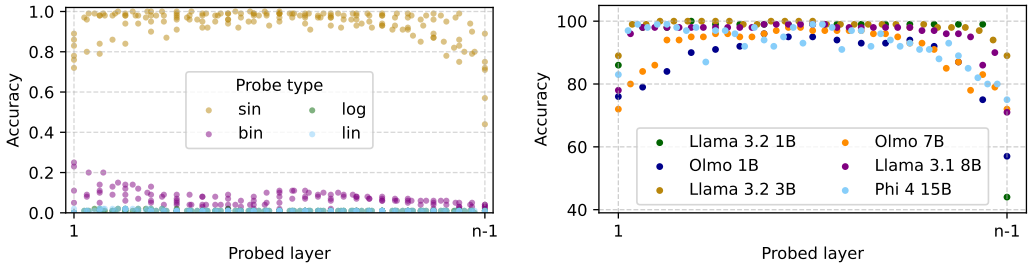
$$\mathbf{S}_{ij} = \begin{cases} \sin(ie^j 1000/d) & \text{if } j \equiv 0 \mod 2 \\ \cos(ie^{j+1} 1000/d) & \text{if } j \equiv 1 \mod 2 \end{cases}$$

where $\mathbf{W}_{\text{in}}$ and $\mathbf{W}_{\text{out}}$ are learned parameters, and $\mathbf{S}$ injects an inductive bias in the classifier towards sinusoidal representations. We build upon an assumption verified by Kadlčík et al. (2025) stating that sinusoidal probes indeed adapt and employ a sinusoidal representation from inductive bias, if a sinusoidal representation is also present in the input embeddings.

We first assess whether sinusoidal probes are the most suitable, i.e., an accurate choice for decoding the *internal* representations of the model. To contextualize the accuracy in terms of sinusoidal quality of the representation, we also evaluate other types of probes used to decode numbers in previous work Feng et al. (2023); Zhu et al. (2025); Kadlčík et al. (2025) as baselines. we train the sinusoidal probe on hidden representations of each layer of Llama 3.2 1B model when processing addition prompts of the form 'x_1 + x_2', where x_1 and x_2 are integers. We specifically evaluate whether the value of x_2 can be retrieved from the model's hidden representation on each layer. To assess generalization rather than memorization capacity, we *split* the prompts and corresponding representations into training, validation and test sets containing *distinct* extracted numbers (x_2).

Figure 3a shows the accuracy of different probes extracting the value of input numbers from representations of each model's layer, evaluated across six different models. The superior accuracy of sinusoidal probe provides an indicator of both (i) the sinusoidal character, and (ii) accuracy of input number representation across layers.

Thus far, an important caveat of ours, as well as a methodology of previous work probing representations of numbers in previous work (section 2) is that we focus on a very specific type of input, namely additions, without any sort of natural language to contextualize them. As such, it is reasonable to assume that the results presented in Figure 3 may not carry to a more natural context: there is little guarantee that the behavior of a model on arithmetic problems is indicative of its behavior a broad range of applications requiring manipulations of numbers in natural language contexts.



(a) using probes with different bases; math contexts (b) across different language models; natural contexts

Figure 3: Accuracy of decoding numeric input token from internal activations of language models

To evaluate the impact of natural language context on accuracy and regularity of numeric representations, we curate a dataset covering four domains involving intensive numeric manipulations: arithmetic (mathematical word problems), temporal (date extraction), medical (ICD-10 diagnostic codes) and culinary (recipe ingredients/quantities). The complete dataset specifications are detailed in Table 1 in Section B. To maximize the quality of learned probes and thus, representativeness of our findings, we also tackle the natural-language class imbalance; as not every number has equal probability to occur in a date or in a recipe, we replace original numbers with integers uniformly sampled from the model vocabulary. We then fit separate probes for activations from every layer, again holding out a set of 100 randomly selected numbers for evaluating generalization.

Fig. 3b displays the accuracies of input value extraction from model's hidden states within natural-language contexts across six LLMs of different sizes and three different families. As we can see, the sinusoidal probes reach over 70% of accuracy in all but three cases and over 90% of accuracy for a majority of probing scenarios. Taking into an account that the learned sinusoidal probe also induce a certain error. Therefore, these accuracies present a lower-bound of accuracy of models' representation of input numbers in natural contexts in a sinusoidal representation — showing that the accurate, sinusoidal representation is indeed characteristic and employed by a wide range of recent language models.

A vast majority of cases performing lower than 80% occurs in probing the models' first and a last layer — we analyse these cases in detail later in Section 3.3.

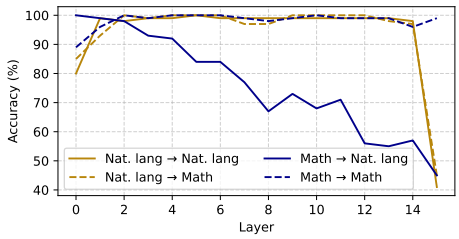


Figure 4: **Generalization of probes** fitted on natural-language occurrences of numeric tokens (solid line), and synthetic, mathematical contexts (dashed line).

A comparison of probes' performance in math an natural-language contexts (Figure 3) may suggest that a type of context does not influence the results of probing accuracy. However, in Figure 4, we show that probes fitted with natural-language contexts are much more *robust* in application across *both* math and natural-language contexts; While both the math and natural-language probes generalize well within their training context type, the natural-language probes are applicable comparably even under a substantial context distribution shift, suggesting that a wider variety of contexts serves as an effective regularization strategy. This observation draws two important implications: First, it informs future work tracing models' internal computations to favor probes trained in natural-language settings. Second, it restrains future work in interpretability from drawing broad conclu-

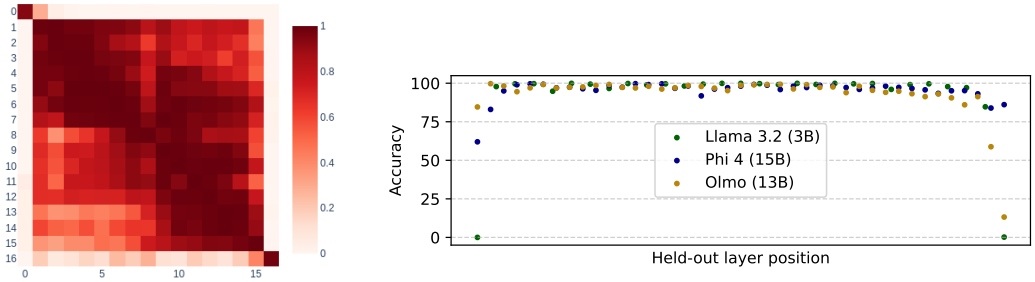
sions on models' mechanics from fully-synthetic settings, evidencing that such conclusions may not generalize to real-world applications.

In short, our results underscore that the same type of sinusoidal representation of numbers holds in general across different model types. Numbers are represented in a similar, systematic fashion, regardless of which layer or type of contexts we consider.

3.3 Different Layers use Interchangeable Numeric Representations

Having found the remarkable degree to which the embeddings for numbers align across different models and the noteworthy universality of sinusoidal representations in varied contexts, we turn to further analyses assessing the extent to which these structures are utilized by the models. First, we address whether the same sinusoidal structure is conserved across a model's computations.

To that end, we employ the probes we developed in Section 3.2 trained separately for each model layer, L_i, and measure the accuracy of each probe on representations derived for every *other* layer $L_j \neq L_i$. This approach, while informative, comes with the caveat that probes trained for a given layer L_i might pick up on idiosyncrasies inherent to a specific layer — i.e., the probes might not disentangle what is specific to numbers as opposed to what is specific to a layer. To address this point, we also fit probes using *all but one* layer $(L_1, L_2, \ldots L_{i-1},\ L_{i+1}, \ldots L_n)$, and evaluate the performance on representations from the held out layer L_i. As previously, we hold out a subset of 100 numbers for assessing generalization in validation and test conditions.



(a) accuracy of probes trained for Llama 1B on a chosen layer (rows) evaluated on all other layers (columns)

(b) probes trained on all-but-one layer, evaluated on the held-out layer

Figure 5: Probes accuracy on activations from unseen *layers*.

Results of probes' cross-layer generalization for Llama 1B are displayed in Figure 5, with largely consistent results for other models in Appendix A.2. We observe that probes fitted on a concrete layer's representations (Figure 5a) generalize outstandingly well to close-by layers, with the exception of the first and the last layer. This trend reveals that language models **operate mostly in a consistent representation** of numbers that is **universal across their computations**, undergoing only minor shifts across layers. This trend is also corroborated by multi-layer probes evaluated in heldout-layer fashion, reaching an accuracy between 95 and 100% across all intermediate layers and three models of diverse sizes and families. This result has practical implications for future work in numeric interpretability — showing that we can train universal, yet highly *accurate* probes for intermediate layers of a broad set of language models[4].

Cross-layer evaluation indicates that models' internal representations tend to differ from the input/output embeddings. We find a notable discrepancy in the *sparsity* of their representation; Whereas the input/output embeddings represent numbers in a more *distributed* fashion, hidden layers use a small number of consistently-ordered sin features.

This is visualized in Figure 6, showing that intermediate layers in Llama 1B lead to probes with very few weights with values $> 10^{-5}$.

[4]We will release our training scripts, together with reproducibility guidelines for training universal probes, in a final version of this paper